NE 723
Fall 2026

Homework Assignment No. 4

Reading Assignment

Nuclear Reactor Theory, Bell & Glasstone, Kriger, 1985

- Chapter 2: Sections 2.5a-2.5d. 2.6.
- Chapter 3: Sections 3.5

Problems

1. Consider a 1D slab ($0 \le x \le a$) with anisotropic scattering and isotropic source $q(x)$. Write down P_N equations and corresponding P_N approximation of the angular flux for
   (a) (5 points) N=3
   (b) (5 points) N=5
2. (10 points)
   Consider a 1D slab ($0 \le x \le a$) with vacuum boundary conditions. Derive Marshak boundary conditions at $x = 0$ and $x = a$ for P_3 method.
3. (10 points)
   Consider a 1D slab ($0 \le x \le a$). Formulate DP_1 method. Derive boundary conditions for DP_1 method in case of
   - vacuum at $x = 0$,
   - reflective boundary at $x = a$.
4. (10 points)
   The P_1 spherical harmonics expansion of the angular flux is given by

$$\psi(\vec{r},\vec{\Omega}) = \phi_{0,0}(\vec{r})Y_{0,0}(\vec{\Omega}) + \phi_{1,-1}(\vec{r})Y_{1,-1}(\vec{\Omega}) + \phi_{1,0}(\vec{r})Y_{1,0}(\vec{\Omega}) + \phi_{1,1}(\vec{r})Y_{1,1}(\vec{\Omega})\,, \tag{1}$$

   where

$$\phi_{l,m}(\vec{r}) = \int_{4\pi} Y^*_{l,m}(\vec{\Omega})\psi(\vec{r},\vec{\Omega})d\vec{\Omega}\,, \tag{2}$$

$$Y_{l,m}(\theta,\gamma) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}P^m_l(\cos\theta)e^{im\gamma}\,. \tag{3}$$

   Derive from Eq. (1) the angular flux in terms of

$$\phi = \int_{4\pi}\psi d\vec{\Omega}\,, \quad J = \int_{4\pi}\vec{\Omega}\psi d\vec{\Omega}\,. \tag{4}$$

Problem	1a	1b	2	3	4	total	%
Points							
Maximum	5	5	10	10	10	40	